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Enrollment ID: DE-2026-0186

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'bad\_bankingdb' selected. The main editor shows a query titled 'Query 1' with the following SQL code:

```
97 FROM account_file
98 WHERE customer_id = 'C001'
99 UNION ALL
100 SELECT 'LOAN FILE' AS source, customer_id, customer_name, address
101 FROM loan_file
102 WHERE customer_id = 'C001'
103 UNION ALL
104 SELECT 'CREDIT CARD FILE' AS source, customer_id, customer_name, address
105 FROM credit_card_file
```

The 'Result Grid' shows the following data:

| source           | customer_id | customer_name | address                  |
|------------------|-------------|---------------|--------------------------|
| ACCOUNT FILE     | C001        | Aung Aung     | No.99, Inya Road, Yangon |
| LOAN FILE        | C001        | Aung Aung     | No.10, Pyay Road, Yangon |
| CREDIT CARD FILE | C001        | Aung Aung     | No.10, Pyay Road, Yangon |

The 'Output' pane shows the 'Action Output' table with the following rows:

| #  | Time     | Action                                                                         | Message                                                             | Duration / Fetch      |
|----|----------|--------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------|
| 30 | 00:18:46 | CREATE TABLE credit_cards ( card_id INT PRIMARY KEY AUTO_INCREMENT, ...        | 0 row(s) affected                                                   | 0.031 sec             |
| 31 | 00:19:18 | SELECT 'ACCOUNT FILE' AS source, customer_id, customer_name, address FROM a... | Error Code: 1146. Table 'good_bankingdb.account_file' doesn't exist | 0.000 sec             |
| 32 | 00:19:25 | USE bad_bankingdb                                                              | 0 row(s) affected                                                   | 0.000 sec             |
| 33 | 00:19:30 | SELECT 'ACCOUNT FILE' AS source, customer_id, customer_name, address FROM a... | 3 row(s) returned                                                   | 0.015 sec / 0.000 sec |

The 'SQLAdditions' pane on the right contains the text: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'bad\_bankingdb' selected. The main editor shows a query titled 'Query 1' with the following SQL code:

```
157
158 }
159
160 CREATE TABLE credit_cards (
161   card_id INT PRIMARY KEY AUTO_INCREMENT,
162   customer_id VARCHAR(20) NOT NULL,
163   card_number VARCHAR(30) NOT NULL UNIQUE,
164   card_type VARCHAR(50),
165   credit_limit DECIMAL(12,2),
166   expiry_date DATE,
167   CONSTRAINT fk_cards_customer
168     FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
169     ON UPDATE CASCADE
170     ON DELETE RESTRICT
171 )
172
```

The 'Output' pane shows the 'Action Output' table with the following rows:

| #  | Time     | Action                                                                  | Message           | Duration / Fetch |
|----|----------|-------------------------------------------------------------------------|-------------------|------------------|
| 26 | 00:11:41 | USE good_bankingdb                                                      | 0 row(s) affected | 0.000 sec        |
| 27 | 00:11:55 | CREATE TABLE customers ( customer_id VARCHAR(20) PRIMARY KEY, custo...  | 0 row(s) affected | 0.015 sec        |
| 28 | 00:17:17 | CREATE TABLE accounts ( account_id INT PRIMARY KEY AUTO_INCREMENT, ...  | 0 row(s) affected | 0.031 sec        |
| 29 | 00:17:55 | CREATE TABLE loans ( loan_id INT PRIMARY KEY AUTO_INCREMENT, custo...   | 0 row(s) affected | 0.031 sec        |
| 30 | 00:18:46 | CREATE TABLE credit_cards ( card_id INT PRIMARY KEY AUTO_INCREMENT, ... | 0 row(s) affected | 0.031 sec        |

The 'SQLAdditions' pane on the right contains the text: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bad\_bankingdb

Tables

account\_file

credit\_card\_file

loan\_file

Views

Stored Procedures

Functions

ecommercedb

good\_bankingdb

Tables

Views

Administration Schemas

Information

Schema: bad\_bankingdb

Query 1 Data redundancy and inconsiste... account\_file credit\_card\_file loan\_file

Limit to 1000 rows

```
217 c.address,
218 a.account_type,
219 a.balance,
220 l.loan_type,
221 l.outstanding_amount,
222 cc.card_number,
223 cc.card_type
224 FROM customers c
225 LEFT JOIN accounts a ON c.customer_id = a.customer_id
226 LEFT JOIN loans l ON c.customer_id = l.customer_id
227 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
228 WHERE c.customer_id = 'C001';
229
230 DROP VIEW IF EXISTS vw_customer_banking_profile;
```

Result Grid

| customer_id | customer_name | address                  | account_type | balance   | loan_type | outstanding_amount | card_number      |
|-------------|---------------|--------------------------|--------------|-----------|-----------|--------------------|------------------|
| C001        | Aung Aung     | No.99, Inya Road, Yangon | Savings      | 500000.00 | Home Loan | 7500000.00         | 4111111111111111 |

Result 3 x

Output

Action Output

# Time Action Message Duration / Fetch

Query Executed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

loan\_file

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Information

Schema: bad\_bankingdb

Query 1 Data redundancy and inconsiste... account\_file credit\_card\_file loan\_file accounts credit\_cards customers loans

Limit to 1000 rows

```
247 cc.card_number,
248 cc.card_type,
249 cc.credit_limit
250 FROM customers c
251 LEFT JOIN accounts a ON c.customer_id = a.customer_id
252 LEFT JOIN loans l ON c.customer_id = l.customer_id
253 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id;
254
255 SELECT * FROM vw_customer_banking_profile WHERE customer_id = 'C001';
```

Result Grid

| customer_id | customer_name | address                  | phone         | account_id | account_type | balance   | loan_id | loan_type |
|-------------|---------------|--------------------------|---------------|------------|--------------|-----------|---------|-----------|
| C001        | Aung Aung     | No.99, Inya Road, Yangon | 09-1111111111 | 1          | Savings      | 500000.00 | 1       | Home Loan |

vw\_customer\_banking\_profile5 x

Output

Action Output

| #  | Time     | Action                                                                         | Message           | Duration / Fetch      |
|----|----------|--------------------------------------------------------------------------------|-------------------|-----------------------|
| 53 | 00:29:57 | SELECT * FROM vw_customer_banking_profile WHERE customer_id = 'C001' LIMIT ... | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 54 | 00:30:07 | SELECT * FROM vw_customer_banking_profile WHERE customer_id = 'C001' LIMIT ... | 1 row(s) returned | 0.016 sec / 0.000 sec |

Query Executed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

loan\_file  
Views  
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ecommercedb  
good\_bankingdb  
Tables  
accounts  
credit\_cards  
customers  
loans  
Views

Administration Schemas

Information

Schema: bad\_bankingdb

Data redundancy and inconsiste... account\_file credit\_card\_file loan\_file accounts credit\_cards customers loans

Limit to 1000 rows

```
-- Verify inconsistency
279
280 SELECT 'ACCOUNT FILE' AS source, address
281 FROM account_file
282 WHERE customer_id = 'C001'
283 UNION ALL
284 SELECT 'LOAN FILE' AS source, address
285 FROM loan_file
286 WHERE customer_id = 'C001'
287 UNION ALL
288 SELECT 'CREDIT CARD FILE' AS source, address
289 FROM credit_card_file
```

Result Grid

| source           | address                  |
|------------------|--------------------------|
| ACCOUNT FILE     | No.99, Inya Road, Yangon |
| LOAN FILE        | No.10, Pyay Road, Yangon |
| CREDIT CARD FILE | No.10, Pyay Road, Yangon |

Result 7 x

Read Only Context Help Snippets

Output

Action Output

| #  | Time     | Action                                                                          | Message                                                  | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------|
| 60 | 00:32:29 | UPDATE account_file SET address = 'No.99, Inya Road, Yangon' WHERE customer_... | 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 | 0.000 sec             |
| 61 | 00:32:37 | SELECT 'ACCOUNT FILE' AS source, address FROM account_file WHERE customer...    | 3 row(s) returned                                        | 0.000 sec / 0.000 sec |

Object Info Session

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

loan\_file  
Views  
Stored Procedures  
Functions  
ecommercedb  
good\_bankingdb  
Tables  
accounts  
credit\_cards  
customers  
loans  
Views

Administration Schemas

Information

Schema: bad\_bankingdb

Data redundancy and inconsiste... accounts credit\_cards customers loans

Limit to 1000 rows

```
306 c.customer_id,
307 c.customer_name,
308 c.address,
309 a.account_type,
310 l.loan_type,
311 cc.card_type
312 FROM customers c
313 LEFT JOIN accounts a ON c.customer_id = a.customer_id
314 LEFT JOIN loans l ON c.customer_id = l.customer_id
315 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
316 WHERE c.customer_id = 'C001';
```

Result Grid

| customer_id | customer_name | address                  | account_type | loan_type | card_type |
|-------------|---------------|--------------------------|--------------|-----------|-----------|
| C001        | Aung Aung     | No.99, Inya Road, Yangon | Savings      | Home Loan | Gold      |

Result 8 x

Read Only Context Help Snippets

Output

Action Output

| #  | Time     | Action                                               | Message           | Duration / Fetch      |
|----|----------|------------------------------------------------------|-------------------|-----------------------|
| 70 | 00:35:01 | SELECT * FROM good_bankingdb.customers LIMIT 0, 1000 | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 71 | 00:35:03 | SELECT * FROM good_bankingdb.loans LIMIT 0, 1000     | 2 row(s) returned | 0.000 sec / 0.000 sec |

Object Info Session